Tier 1 roadmap (SCOUT / VECTOR) — living checklist

Location: `sports/documents/tier_1_roadmap.md` (alongside other sports-facing notes; not a duplicate of theory memos in `5-Manuscript`.)

Notebook: `sports/535_sports_tier_1.ipynb`

Alex-sequenced modeling notebook: `sports/538_alex_tier1_model_and_fit.ipynb` (mirrors `5-Manuscript/Alex_Tier1_Sequential_Model_Outline.md`; `537` remains the simulation-only lab).

Purpose: Single place for the mechanical pipeline, the `df` column contract after each stage, and **CELL 0** switches. Update this file as CELL 4+ and modeling land.

Contract (what "must be true" for implementation): this file + the **module docstrings / function contracts** in `sports_pipeline/tier1_mechanism_vars.py` and `sports_pipeline/panel_build.py` (especially LOO / `poolq_loo` behavior). Manuscripts in `5-Manuscript` are **design intent and exposition**, not the execution contract unless we explicitly mirror a rule here.

Related theory (`5-Manuscript`, stay at source — link, don't copy):

- `Vector_to_Scout_Tier1_Modeling_Direction.md` — main Tier 1 modeling direction.
- `tier_1_model.md` — model layer notes.
- `Tier1_Briefing_Outline.md` — outline answering Alex's May 5 briefing guidance: model components, assumptions, fitting plan, and turning point.
- `Alex_Tier1_Sequential_Model_Outline.md` — **advisor-order spine** (minimal model → data → fit → `L^*`); links here and to `538_alex_tier1_model_and_fit.ipynb`.
- `Tier1_Narrative_Outline.md` — current narrative flow: competing local forces, L-first Tier 1 model, Λ, fitting plan, variable domains, and reference addenda.
- `2026_0430_Paper7_feedback.md` — theory vs minimal model; scarcity.

Optional depth (broader VECTOR notes): `Vector_Master_Theory_and_Modeling_Notes*.md`, `Vector_Questions_and_Modeling_Thoughts.md`, `Vector_Evans_Reaction_and_Theoretical_Expansion_Notes.md` — pull excerpts into this roadmap only when a line becomes a **coded rule** or sample definition.

Where We Have Been / Where We Are Going

Keep this section open while working in `535_sports_tier_1.ipynb`. Green items are done/validated; black/default items are next work or open decisions.

Done / Validated

- DONE — Mechanical notebook structure: `535_sports_tier_1.ipynb` now follows the named-cell pattern: markdown header, then code cell with matching first comment line.
- DONE — CELL 0 switchboard: run gates and main knobs live in CELL 0 (`RUN_CELL1`–`RUN_CELL4`, `PERF_METRIC`, `PRIMARY_POOL_MODE`, `COMPUTE_WEIGHTED_CROWDING`, etc.).
- DONE — 530 source-of-truth workflow: exact 530 reproduction requires running `530_sports_pipeline.ipynb` from scratch with `use_prebuilt_panel_csv=False` and `export_panel_after_run=True`, then loading the refreshed

`datasets/mbb/player_season_panel_530.csv` in 535.

- DONE — Reproduced the reduced-form phenomenon: with the refreshed 530 panel, `poolq_loo` binned bars reproduce the inverted-U / top-bin-drop pattern.
- DONE — Plot-style switch: CELL 4 supports `CELL4_EDA_PLOT_STYLE = "poolq_line"` vs `"bins_bars_520"` so we can compare quality-scale lines with 530-style bin-index bars.
- DONE — Tier 1 mechanism variables: CELL 3 creates `congestion_quality`, `congestion_crowding`, and, when enabled, `congestion_crowding_weighted`.
- DONE — Weighted crowding behavior made explicit: when `PRIMARY_POOL_MODE = "crowding"` and `COMPUTE_WEIGHTED_CROWDING = True`, CELL 4 plots `congestion_crowding_weighted`; when the boolean is false, it plots plain `congestion_crowding`.
- DONE — First Tier 1 EDA pass: binned bars for `congestion_quality`, plain `congestion_crowding`, and weighted crowding have been compared against the original `poolq_loo` pattern.

Next / TODO

- TODO — Decide the first empirical proxy for L_{ijt}: likely start with `poolq_loo` / `congestion_quality`; treat `congestion_crowding`, `congestion_crowding_weighted`, and `minutes` as later decomposition diagnostics unless they earn a clearer first-pass role.
- TODO — Fix or demote the right-panel dashed model curve: current clipped LPM / PD-style curves can be misleading; print unclipped predictions and coefficients before trusting the curve.
- TODO — Add a cleaner CELL 5 regression block: start with $Y \sim L + L^2 + A$ where L is first proxied by `poolq_loo` / `congestion_quality`; then add one decomposition diagnostic at a time (`congestion_crowding`, weighted crowding, `minutes`, interactions) after the minimal shape is clear.
- TODO — Decide binning defaults by purpose: use `equal_width` when matching a 530 export slug; use `quantile` first for Tier 1 mechanism diagnostics so bins are populated.
- TODO — Add export behavior for 535 figures/tables only after the Tier 1 specs settle enough to save reproducible outputs.
- TODO — Update theory language after EDA: distinguish reduced-form `poolq_loo` from mechanism variables (`quality`, `crowding`, weighted crowding).

Pipeline (numbered)

1. **Load the panel** — `load_panel(CFG)` → `df` from `player_season_panel_530.csv` at **workspace root** `datasets/mbb/` (via `sports_pipeline.paths.panel_530_csv()`). For exact 530 reproduction, refresh this CSV first by running **530** from scratch with `use_prebuilt_panel_csv=False` and `export_panel_after_run=True`.

2. **Set `perf` from a metric** — e.g. PPM via `assign_perf_from_metric(df, PERF_METRIC)` with `PERF_METRIC` defined in **CELL 0**.

3. **Recompute legacy `poolq_loo` (and `poolq_sq`)** — `recompute_teammate_loo_pool_quality(...)`.

   Why recompute instead of trusting the CSV?

   - LOO pool quality must be built off the **`perf` you just chose**; a saved `poolq_loo` may reflect an older or different `perf` definition.
   - Optional **winsor** on `poolq_loo` via `CFG.poolq_winsor_quantiles` is applied in this step when set.

   Comparison: `poolq_loo` (legacy) and **`congestion_quality`** (Tier 1) can differ slightly when some roster rows have missing `perf`; the Tier 1 helper uses only rows with valid `perf` in the `(team_id, season)` group for denominators. See `sports_pipeline/tier1_mechanism_vars.py` module docstring.

4. **Add Tier 1 mechanism columns** — `add_tier1_mechanism_variables(...)`:

   - `congestion_quality` — LOO **mean** teammate `perf` (valid Q for Tier 1).
   - `congestion_crowding` — LOO **sum** teammate `perf`.
   - `congestion_crowding_weighted` — optional weighted variant (minutes × `perf` logic); gated by **CELL 0** `COMPUTE_WEIGHTED_CROWDING`.
   - If `MIN_MINUTES_TIER1` is set in **CELL 0**, rows below that **minutes** threshold are dropped **before** group stats.

5. **CELL 4 — Tier 1 EDA** (notebook): On a sample with valid **`PRIMARY_POOL_MODE`** pool column, **`congestion_crowding`**, **`minutes`**, **`Y_draft`**: descriptive `describe()`; **binned** mean draft vs Q using **`assign_poolq_bin_labels`** with **`CFG.ventiles`** and **`CFG.poolq_binning`**; numpy **LPM** `Y ~ 1 + Q + Q²`; multivariate linear `Y ~ 1 + Q + Q² + C + minutes` when Q is quality (if Q is crowding, **omit duplicate C**); **PD-style** predicted curve over a Q grid with C and minutes at sample medians (linear approx, clipped to [0,1]). When `PRIMARY_POOL_MODE="crowding"` and `COMPUTE_WEIGHTED_CROWDING=True`, the binned Q variable is `congestion_crowding_weighted`; otherwise it is plain `congestion_crowding`.

`df` column contract (Tier 1–relevant)

After steps 1–3 (after notebook CELL 2):

- Required for assertions in the notebook: `Y_draft`, `poolq_loo`, `minutes`.
- Always present after recompute (when inputs allow): **`perf`**, **`poolq_loo`**, **`poolq_sq`**.
- Plus **all other columns** from the 530 CSV (identifiers such as `athlete_id`, `team_id`, `season`, and any merges from 530).

After step 4 (after notebook CELL 3):

- Same as above, **plus**: `congestion_quality`, `congestion_crowding`, and usually `congestion_crowding_weighted` (or NaNs if weighted path is off / invalid inputs).

CELL 0 — variables / boolean switches (current)

Kind	Name	Role
Run gate	`RUN_CELL1`	Environment · imports · panel path
Run gate	`RUN_CELL2`	Load panel · PPM · legacy `poolq_loo`
Run gate	`RUN_CELL3`	Tier 1 mechanism variables
Run gate	`RUN_CELL4`	Tier 1 EDA (bins, LPM, PDP-style Q grid)
Knob	`PERF_METRIC`	Passed to `assign_perf_from_metric` (default `"ppm"`)
Knob	`CELL4_MATCH_530_VENTILE`	`True` = 530-style `poolq_loo`; `False` = Tier 1 mechanism variable path
Knob	`CELL4_EDA_PLOT_STYLE`	`"poolq_line"` vs `"bins_bars_520"`
Knob	`MIN_MINUTES_TIER1`	Optional row filter before Tier 1 group stats; `None` = full sample
Knob	`PRIMARY_POOL_MODE`	`"quality"` vs `"crowding"` for Tier 1 CELL 4 path
Knob	`COMPUTE_WEIGHTED_CROWDING`	Whether to build `congestion_crowding_weighted`; when `PRIMARY_POOL_MODE="crowding"`, `True` makes CELL 4 plot weighted crowding

Pattern matches `540_tenure_pipeline.ipynb` CELL 0: one switchboard, re-run after changes.

Next (placeholder)

- **CELL 5+:** proper logit / `statsmodels`; cluster SEs; season or team FE; richer PDP; export figures to `CFG.exports_dir` if you want parity with **530** PNG exports.

Changelog

Date	Note
2026-05-12	Added `Alex_Tier1_Sequential_Model_Outline.md` and bootstrap `538_alex_tier1_model_and_fit.ipynb` for Alex-ordered exposition; `537` unchanged (simulation).
2026-05-07	Added `Tier1_Narrative_Outline.md` to related theory and updated TODOs to reflect the current L-first Tier 1 framing with crowding/minutes as decomposition diagnostics.
2026-05-06	Added manuscript briefing outline link for Alex's May 5 guidance: model components, assumptions, fitting plan, and Q^* turning point.
2026-05-05	Added "Where We Have Been / Where We Are Going" dashboard; documented 530 source-of-truth workflow and weighted-crowding CELL 4 behavior.
2026-05-06	**CELL 4** in 535: binned draft vs Q, quadratic LPM, C + minutes controls, PDP-style Q grid; `RUN_CELL4` in CELL 0.
2026-05-06	`paths.project_root` = Ivy_Net workspace root so `datasets/mbb/` matches 530 output (not `sports/datasets/mbb/`).
2026-05-05	File lives under `sports/documents/`; contract vs theory section + links to `5-Manuscript`.
2026-03-31	Initial file: pipeline steps, recompute rationale, `df` contract, CELL 0 table.